

Question 1

P1

Youyi wants to buy a new cell phone from a company that offers 33 different types of cell phones. Each cell phone is available in 5 different colours. Determine the total number of available options.

Solution

$$\underline{33} \bullet \underline{5} = 165 \text{ options}$$

1 mark

Question 2

T1

Determine the central angle subtended by an arc length of 30 cm on a circle with a diameter of 20.6 cm. Express your answer in degrees.

Solution

$$s = \theta r$$

1 mark for substitution

$$30 = \theta(10.3)$$

$$\theta = 2.912\ 621\dots$$

$$\theta = (2.912\ 621)\left(\frac{180^\circ}{\pi}\right)$$

1 mark for conversion

$$\theta = 166.880\ 911\dots^\circ$$

$$\theta = 166.881^\circ$$

2 marks

Question 3

P3

A choir has 14 sopranos. Karin and Ellen are two of them. Determine the number of possible ensembles of 3 sopranos that could be formed if either Karin or Ellen, or both, must be included.

Solution

Method 1

$$\text{Case 1: } {}_1C_1 \cdot {}_{12}C_2 = 66$$

1 mark for Case 1 and Case 2

$$\text{Case 2: } {}_1C_1 \cdot {}_{12}C_2 = 66$$

$$\text{Case 3: } {}_2C_2 \cdot {}_{12}C_1 = 12$$

1 mark for Case 3

$$66 + 66 + 12 = 144$$

1 mark for addition of cases

3 marks

Method 2

$${}_{14}C_3 - {}_{12}C_3 = 144$$

1 mark for ${}_{14}C_3$

1 mark for ${}_{12}C_3$

1 mark for subtraction of cases

3 marks

Question 4

R10

Loïc invests \$5000 at an annual interest rate of 6.75% compounded semi-annually. The value of his investment grows according to the following formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A is the value of the investment after t years

P is the amount of the initial deposit

r is the annual interest rate (as a decimal)

n is the number of compounding periods per year

t is the time in years

Determine, algebraically, the length of time it will take Loïc's investment to grow to \$9500.

Solution

$$9500 = 5000 \left(1 + \frac{0.0675}{2} \right)^{2t} \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.9 = (1.03375)^{2t}$$

$$\log 1.9 = \log(1.03375)$$

1 mark for applying logarithms

$$\log 1.9 = 2t \log 1.03375$$

1 mark for power law (or appropriate logarithm laws)

$$t = \frac{\log 1.9}{2 \log 1.03375}$$

$$t = 9.668\ 522\dots$$

$\frac{1}{2}$ mark for evaluating a quotient of logarithms

$$t = 9.669 \text{ years}$$

$$\therefore t = 10 \text{ years}$$

3 marks

Note:

Accept $t = 9.669$ years as a final answer (or consistent value of t).

Question 5

T5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta - \csc \theta - 6 = 0$$

Solution

Method 1

$$(\csc \theta - 3)(\csc \theta + 2) = 0$$

$$\csc \theta = 3$$

$$\csc \theta = -2$$

1 mark for solving for $\csc \theta$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for reciprocal

$$\theta_r = 0.339\ 836\dots$$

$$\theta = 0.340; 2.802$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Method 2

$$\frac{1}{\sin^2 \theta} - \frac{1}{\sin \theta} - 6 = 0$$

1 mark for reciprocal

$$1 - \sin \theta - 6 \sin^2 \theta = 0$$

$$6 \sin^2 \theta + \sin \theta - 1 = 0$$

$$(3 \sin \theta - 1)(2 \sin \theta + 1) = 0$$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for solving for $\sin \theta$

$$\theta_r = 0.339\ 836\dots$$

$$\theta = 0.340; 2.802$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Question 6

P1

Determine how many 4-digit numbers can be made using the digits 0, 1, 2, 3, 4, and 5, if the third digit must be 3, and repetition is not allowed.

Solution

$$\frac{4}{\text{not 0 or 3}} \cdot \frac{4}{\quad} \cdot \frac{1}{3} \cdot \frac{3}{\quad} = 48$$

½ mark for restriction of first digit

½ mark for restriction of third digit

1 mark for fundamental counting principle

2 marks

Question 7

R3, R5

Describe the transformations required to obtain the graph of the function $y = -4f\left(\frac{1}{3}x\right)$ from the graph of $y = f(x)$.

Solution

- Reflection over the x -axis
- Vertical stretch by a factor of 4
- Horizontal stretch by a factor of 3

1 mark for vertical reflection

1 mark for vertical stretch

1 mark for horizontal stretch

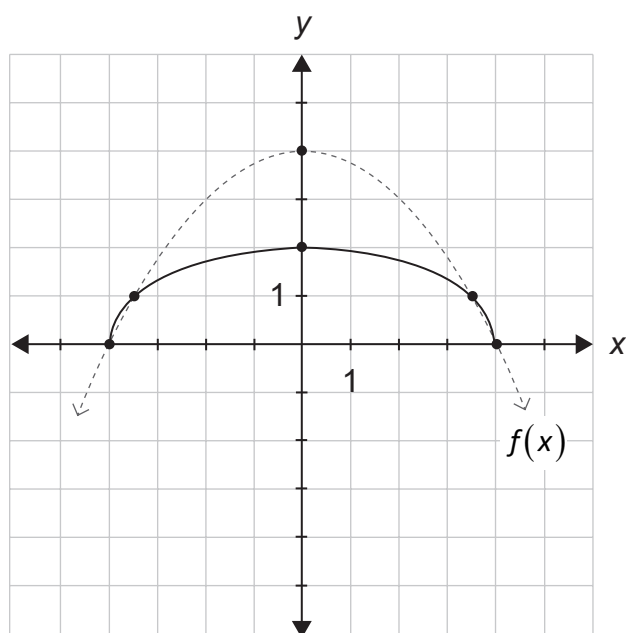
3 marks

Question 8

R13

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points,
 $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points,
 $\{1 \leq y \leq 2\}$

2 marks

Question 9

R11

Describe how to determine if $(x - a)$ is a factor of the polynomial function, $p(x)$.

Solution

$(x - a)$ is a factor of $p(x)$, if and only if $p(a) = 0$.

1 mark

Question 10

R8

Express $\frac{1}{2}\log x - 2\log y + \log z$ as a single logarithm.

Solution

$$\log\left(\frac{\sqrt{x} \cdot z}{y^2}\right)$$

1 mark for product law

1 mark for quotient law

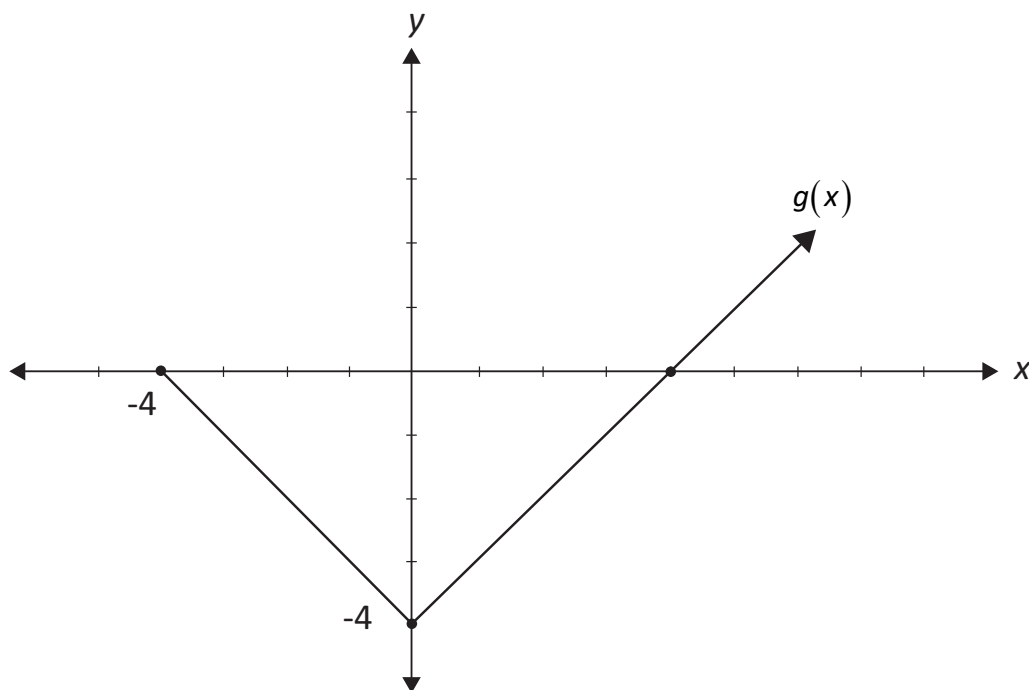
1 mark for power law ($\frac{1}{2}$ mark for each)

3 marks

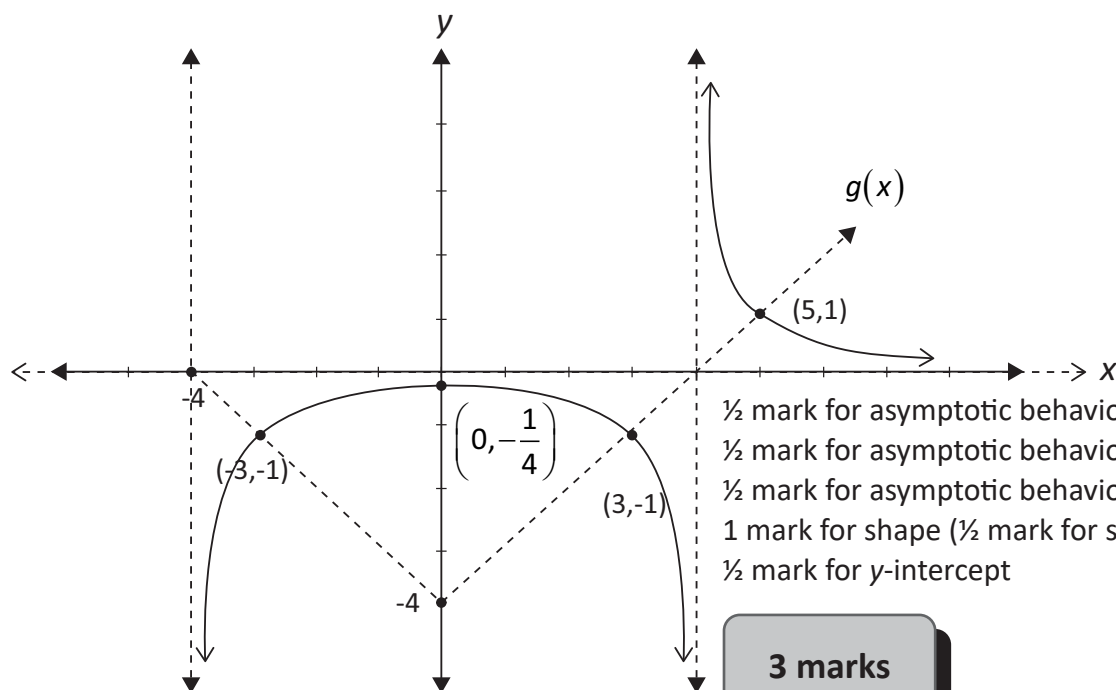
Question 11

R1

Given the graph of $g(x)$, sketch the graph of $f(x) = \frac{1}{g(x)}$.



Solution



$\frac{1}{2}$ mark for asymptotic behaviour approaching $x = -4$
 $\frac{1}{2}$ mark for asymptotic behaviour approaching $x = 4$
 $\frac{1}{2}$ mark for asymptotic behaviour approaching $y = 0$
 1 mark for shape ($\frac{1}{2}$ mark for shape in each section)
 $\frac{1}{2}$ mark for y-intercept

3 marks

Question 12

T6

Given the identity $\frac{1 - \tan^2 \theta}{\sec^2 \theta} = \cos 2\theta$,

- state a non-permissible value of θ .
- prove the identity for all permissible values of θ .

Solution

a) Any value where $\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$ or $\theta = 90^\circ + 180^\circ k, k \in \mathbb{Z}$, is acceptable.

1 mark

b)

Method 1

Left-Hand Side	Right-Hand Side
$ \begin{array}{r} 1 - \frac{\sin^2 \theta}{\cos^2 \theta} \\ \hline \frac{1}{\cos^2 \theta} \\ \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \\ \hline \frac{1}{\cos^2 \theta} \\ \frac{\cos 2\theta}{\cancel{\cos^2 \theta}} \\ \hline \frac{1}{\cancel{\cos^2 \theta}} \\ \cos 2\theta \end{array} $	$\cos 2\theta$

1 mark for correct substitution of identities

1 mark for algebraic strategies

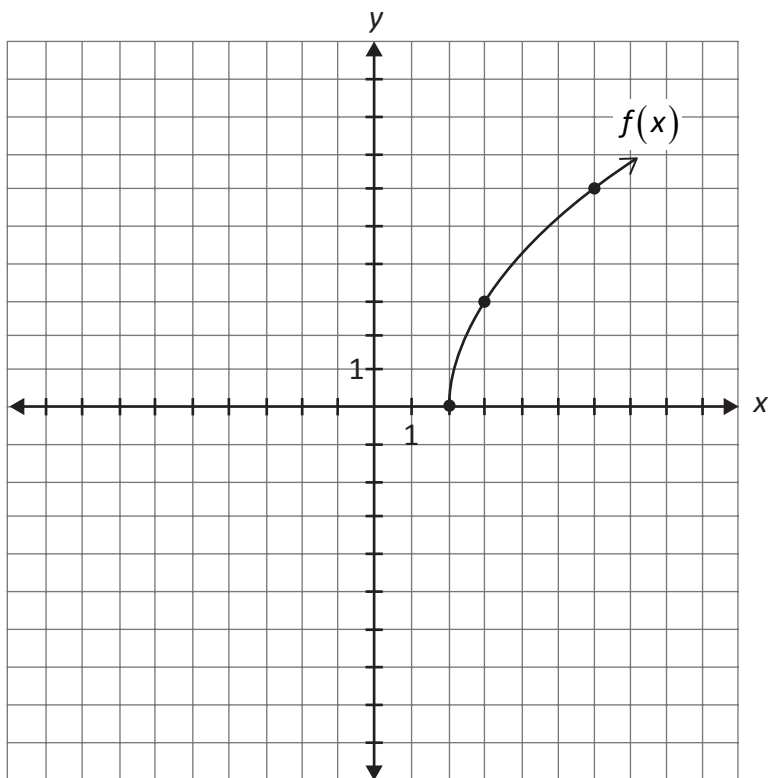
1 mark for logical process to prove the identity

3 marks

Question 13

R13

Determine an equation of the radical function represented by the graph of $f(x)$.



Solution

$$f(x) = \underline{3\sqrt{x-2}}$$

1 mark for vertical stretch

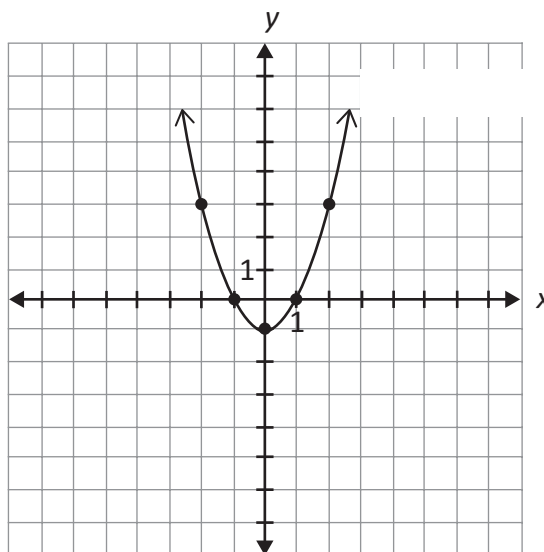
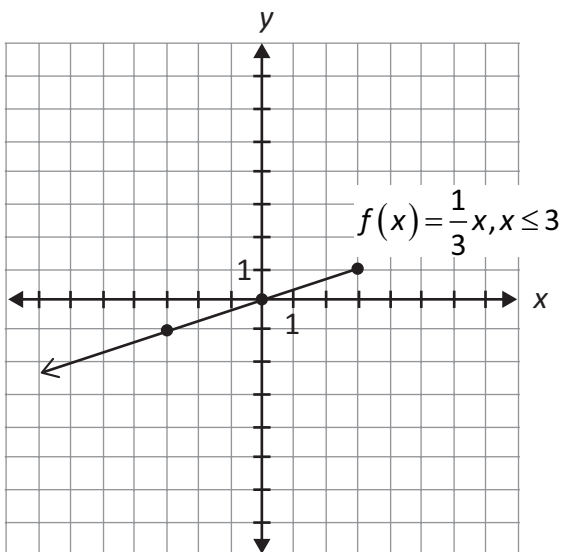
1 mark for horizontal translation

2 marks

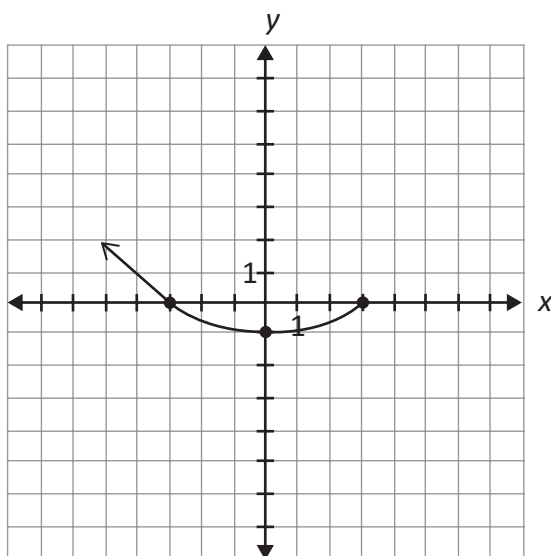
Question 14

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(f(x))$.



Solution



1 mark for $g(f(x))$

1 mark for restricted domain

2 marks

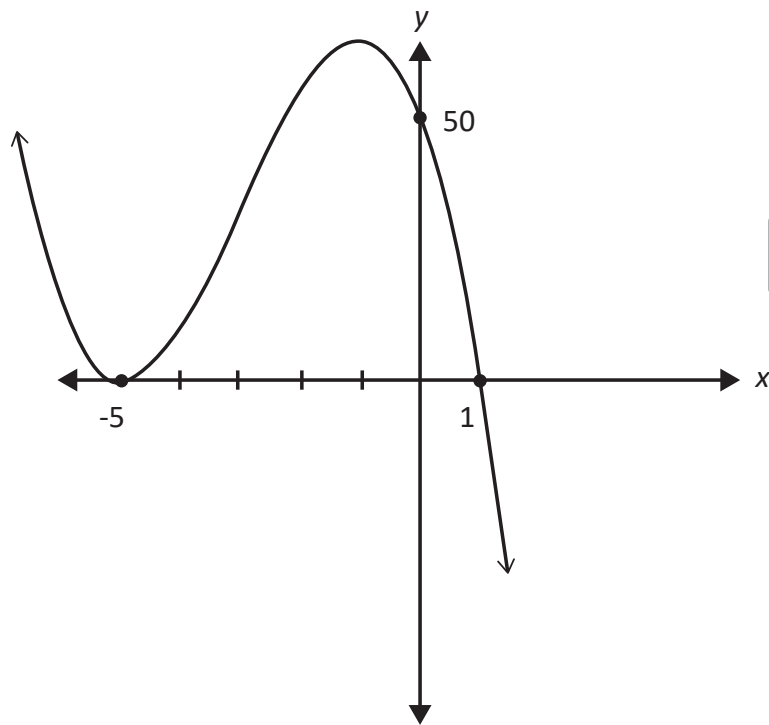
Question 15

R12

Sketch a graph of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 3
- $p(x)$ has a zero at -5 with a multiplicity of 2
- $p(x)$ has a zero at 1
- $p(x)$ has a leading coefficient of -2

Solution



$\frac{1}{2}$ mark for x-intercepts
1 mark for multiplicity of 2 at $x = -5$
 $\frac{1}{2}$ mark for y-intercept
 $\frac{1}{2}$ mark for end behaviour
 $\frac{1}{2}$ mark for shape of cubic function

3 marks

Question 16

R14

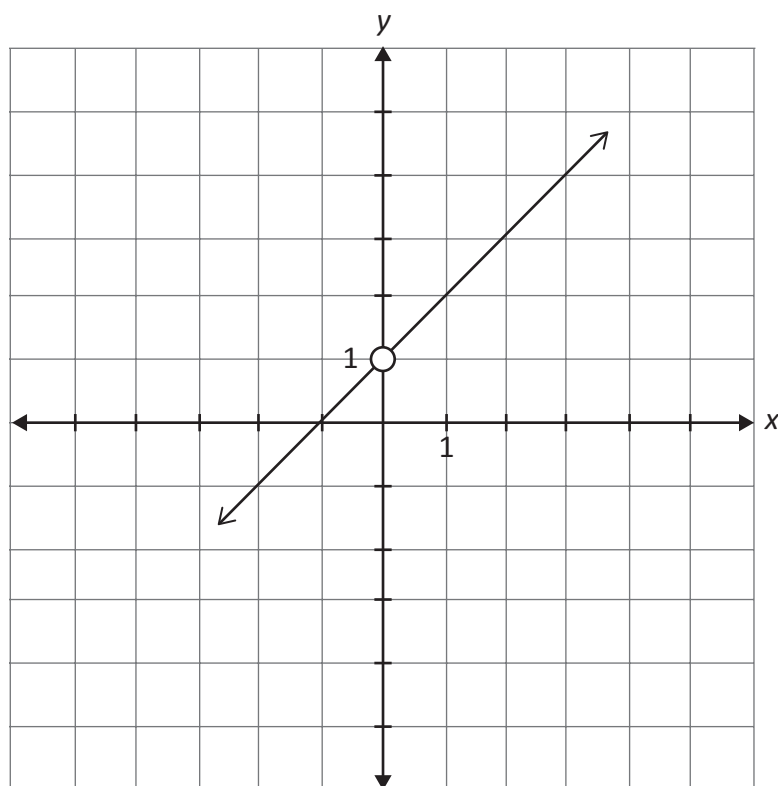
Sketch the graph of $f(x) = \frac{x^2 + x}{x}$.

Solution

$$f(x) = \frac{\cancel{x}(x+1)}{\cancel{x}}$$

$$f(x) = x + 1, x \neq 0$$

$\therefore$ there is a point of discontinuity (hole) at $(0,1)$.



1 mark for point of discontinuity (hole) at $x = 0$

1 mark for shape of $y = x + 1$

2 marks

Question 17

R2

Identify the function that has a different range than $f(x) = x^2$.

a. $y = 4f(x)$

b. $y = f(x) + 4$

c. $y = f(x + 4)$

d. $y = f(-x)$

Question 18

T3

Identify the exact value of $\sin\left(-\frac{7\pi}{3}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. $-\frac{1}{2}$

c. $\frac{1}{2}$

d. $\frac{\sqrt{3}}{2}$

Question 19

R7

Identify the value of a in the equation, $\log_a 3 = \frac{1}{4}$.

a. $\frac{1}{64}$

b. $3^{\frac{1}{4}}$

c. 12

d. 81

Question 20**T3**

Identify a value of x where the graphs of $y = \cos x$ and $y = \sin x$ intersect.

a. $-\frac{3\pi}{4}$

b. $-\frac{\pi}{4}$

c. 0

d. $\frac{3\pi}{4}$

Question 21**P4**

Identify the number of negative terms in the binomial expansion of $(2x - y)^6$.

a. 0

b. 3

c. 4

d. 6

Question 22**T5**

Identify the general solution of the equation, $\sin x = 1$.

a. $x = k\pi, k \in \mathbb{Z}$

b. $x = 2k\pi, k \in \mathbb{Z}$

c. $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

d. $x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

Question 23**R12**

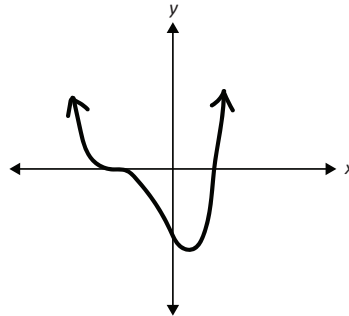
Identify a possible degree of the polynomial function represented below.

a. 2

b. 3

c. 4

d. 5

**Question 24****T6**

Identify the expression that is equivalent to $\cos x \cot x$.

a. $\frac{1 - \sin x}{\sin x}$

b. $\frac{\sin^2 x - 1}{\sin x}$

c. $\sin x$

d. $\frac{1 - \sin^2 x}{\sin x}$

Question 25**R1**

Given that the point $(-7, -5)$ is on the graph of $y = g(x)$, identify the corresponding point on the graph of $y = |g(x)|$.

a. $(7, 5)$

b. $(-7, 5)$

c. $(7, -5)$

d. $(-5, -7)$

Question 26

R11

Given that $x = 1$ is one of the zeros of $p(x) = x^3 + x^2 - 10x + 8$, express $p(x)$ in completely factored form.

Solution

$$p(x) = x^3 + x^2 - 10x + 8$$

1		1	1	-10	8
		↓	1	2	-8
		1	2	-8	0

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x - 1)(x^2 + 2x - 8)$$

$$p(x) = (x - 1)(x + 4)(x - 2)$$

1 mark for consistent factors

2 marks

Question 27

T3

Determine the exact value of $\frac{\csc^2 \theta - 1}{1 - \cos^2 \theta}$ if $\theta = \frac{5\pi}{6}$.

Solution

Method 1

$$\frac{\csc^2\left(\frac{5\pi}{6}\right) - 1}{1 - \cos^2\left(\frac{5\pi}{6}\right)}$$

$$\frac{(2)^2 - 1}{1 - \left(-\frac{\sqrt{3}}{2}\right)^2}$$

$$\frac{4 - 1}{1 - \frac{3}{4}}$$

$$\frac{3}{\frac{1}{4}}$$

$$12$$

1 mark for $\csc\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

1 mark for $\cos\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

2 marks

Method 2

$$\frac{\cot^2 \theta}{\sin^2 \theta}$$

$$\frac{\cot^2\left(\frac{5\pi}{6}\right)}{\sin^2\left(\frac{5\pi}{6}\right)}$$

$$\frac{(-\sqrt{3})^2}{\left(\frac{1}{2}\right)^2}$$

$$\frac{3}{\frac{1}{4}}$$

$$12$$

1 mark for $\cot\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

1 mark for $\sin\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

2 marks

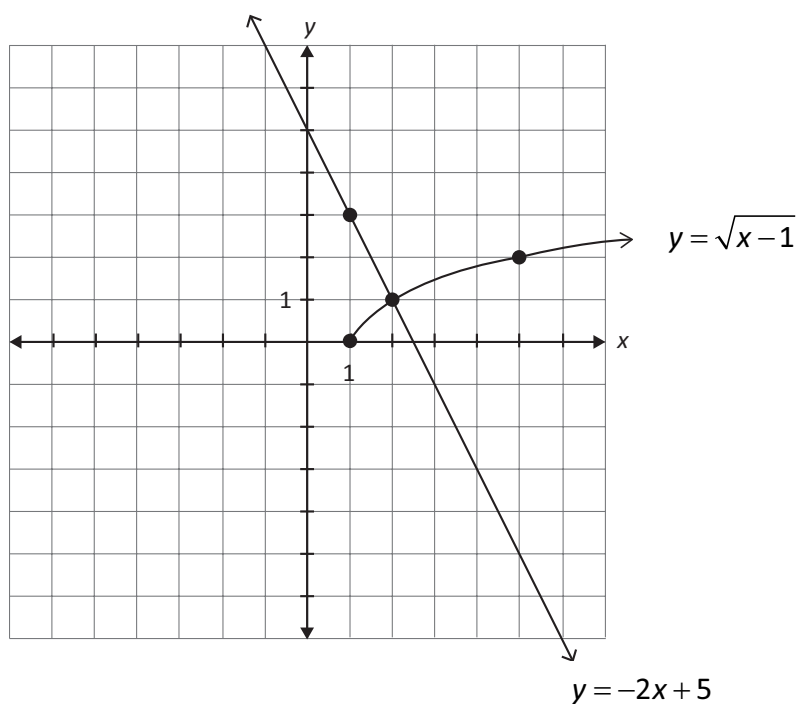
Question 28

R13

Solve, graphically.

$$-2x + 5 = \sqrt{x - 1}$$

Solution



$$\therefore x = 2$$

1 mark for shape of a radical function

1 mark for horizontal translation

1 mark for x-value consistent with intersection point of the two graphs

3 marks

Question 29

R8

If $\log_a 2 = p$ and $\log_a 5 = q$, express $\log_a 40$ in terms of p and q .

Solution

$$\begin{aligned}\log_a 40 &= \log_a (2^3 \cdot 5) \\ &= 3\log_a 2 + \log_a 5 \\ &= 3p + q\end{aligned}$$

1 mark for power law

1 mark for product law

2 marks

Question 30

R6

Describe how to sketch the graph of $y=f^{-1}(x)$, given the graph of a linear function, $y=f(x)$.

Solution

The graph of $y=f^{-1}(x)$ is obtained by reflecting the graph of $y=f(x)$ over the line $y=x$.

or

To sketch the inverse, $y=f^{-1}(x)$, transform all points on the graph of $y=f(x)$ from (x,y) to (y,x) .

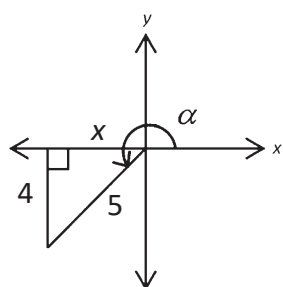
1 mark

Question 31

T6

Given that $\sin \alpha = -\frac{4}{5}$ where α is in quadrant III, and $\cos \beta = \frac{12}{13}$ where β is in quadrant IV, determine the exact value of $\cos(\alpha - \beta)$.

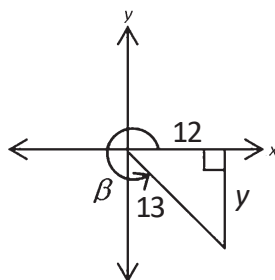
Solution



$$x^2 + 4^2 = 5^2$$

$$x^2 = 9$$

$$x = \pm 3$$



$$12^2 + y^2 = 13^2$$

$$y^2 = 25$$

$$y = \pm 5$$

½ mark for value of x

½ mark for value of y

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \left(-\frac{3}{5}\right)\left(\frac{12}{13}\right) + \left(-\frac{4}{5}\right)\left(-\frac{5}{13}\right)$$

$$= \frac{-36 + 20}{65}$$

$$= -\frac{16}{65}$$

½ mark for consistent value of $\cos \alpha$

½ mark for consistent value of $\sin \beta$

1 mark for substitution into correct identity

3 marks

Note:

Accept any of the following values for x: $x = \pm 3$; $x = -3$; or $x = 3$.

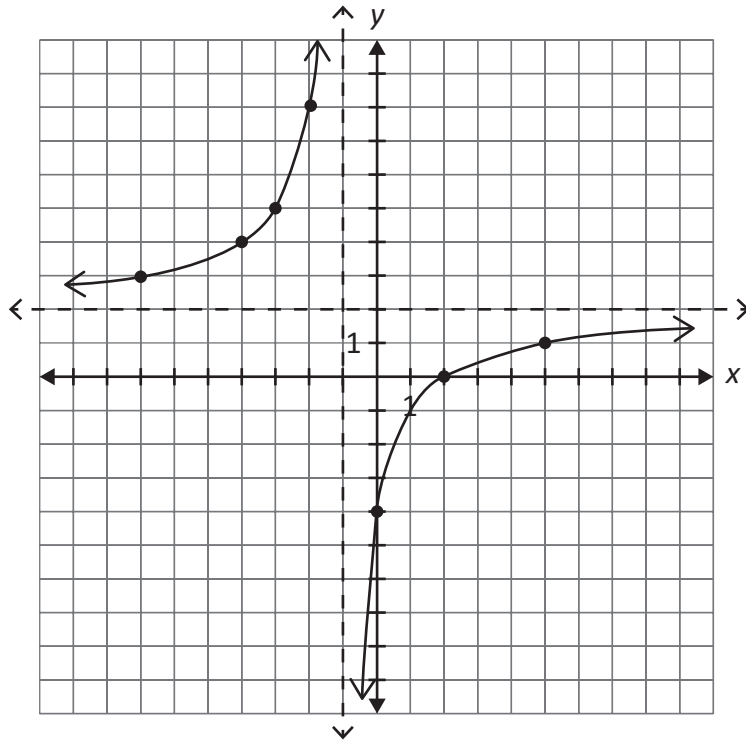
Accept any of the following values for y: $y = \pm 5$; $y = -5$; or $y = 5$.

Question 32

R14

Sketch the graph of $f(x) = \frac{2x-4}{x+1}$.

Solution



1 mark for asymptotic behaviour approaching $x = -1$

1 mark for asymptotic behaviour approaching $y = 2$

1 mark for shape of rational function

(½ mark for graph left of $x = -1$, ½ mark for graph right of $x = -1$)

3 marks

Question 33

T3, T4

Due to the tidal variations, the depth of the water in a bay is represented by the function,

$$d(t) = -2\cos\left[\frac{\pi}{6}(t+4)\right] + 16$$

where d is the depth of the water, in metres and
 t is the time, in hours.

- a) Determine the depth of the water when $t = 2$.
- b) State the minimum depth of the water.
- c) Determine the length of time between two consecutive low tides (minimum depths).

Solution

$$\begin{aligned}\text{a) } d(2) &= -2\cos\left[\frac{\pi}{6}(2+4)\right] + 16 \\ &= -2\cos\pi + 16 \\ &= -2(-1) + 16 \\ &= 18 \text{ metres}\end{aligned}$$

1 mark for exact value

1 mark

b) 14 metres

1 mark

$$\begin{aligned}\text{c) period} &= \frac{2\pi}{\frac{\pi}{6}} \\ &= 12 \text{ hours}\end{aligned}$$

1 mark for period

1 mark

Question 34

T3

Evaluate.

$$\sin^2\left(\frac{5\pi}{4}\right)\sec\left(\frac{13\pi}{3}\right)-\cot\left(\frac{7\pi}{4}\right)$$

Solution

$$\left(-\frac{\sqrt{2}}{2}\right)^2\left(\frac{2}{1}\right)-(-1)$$

$$\frac{1+1}{2}$$

1 mark for $\sin\left(\frac{5\pi}{4}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\sec\left(\frac{13\pi}{3}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\cot\left(\frac{7\pi}{4}\right)$ (½ mark for value; ½ mark for quadrant)

3 marks

Question 35

R14

a) State the non-permissible value(s) of the function, $f(x) = \frac{x-3}{x^2-9}$.

b) Describe what each non-permissible value represents on the graph of $f(x) = \frac{x-3}{x^2-9}$.

Solution

a) $x = 3, -3$

1 mark for non-permissible values ($\frac{1}{2}$ mark for each)

1 mark

b) At $x = 3$, there is a point of discontinuity (hole) on the graph of $f(x)$. At $x = -3$, there is a vertical asymptote on the graph of $f(x)$.

$\frac{1}{2}$ mark for point of discontinuity (hole) at $x = 3$

$\frac{1}{2}$ mark for vertical asymptote at $x = -3$

1 mark

Question 36

R7

Determine which of the given values is greater. Justify your answer.

$$\log_3 38 \text{ or } \log_4 50$$

Solution

$$3^3 = 27$$

$$3^4 = 81 \quad \log_3 38 > 3$$

$$4^2 = 16$$

$$4^3 = 64 \quad \log_4 50 < 3$$

$\therefore \log_3 38$ is greater

1 mark for justification

1 mark

Determine if the point $\left(\frac{3}{4}, \frac{\sqrt{5}}{4}\right)$ is on the unit circle. Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\text{LHS} = \left(\frac{3}{4}\right)^2 + \left(\frac{\sqrt{5}}{4}\right)^2$$

$$= \frac{9}{16} + \frac{5}{16}$$

$$= \frac{14}{16}$$

$$\frac{14}{16} \neq 1$$

$\therefore$ The point is not on the unit circle.

1 mark for justification

1 mark

Question 38

P4

Determine which term contains x^5 in the binomial expansion of $\left(2x + \frac{3}{x}\right)^{11}$.

Solution

Method 1

$$x^{11}, (x^{10})\left(\frac{1}{x}\right), (x^9)\left(\frac{1}{x^2}\right), \dots$$

1 mark for determining a pattern

$$x^{11}, x^9, x^7, \dots$$

$\therefore$ The fourth term contains x^5 .

1 mark for fourth term
(or a term consistent with the pattern)

2 marks

Method 2

$$(x)^{11-k} (x^{-1})^k = x^5$$

$\frac{1}{2}$ mark for substitution

$$x^{11-2k} = x^5$$

$$11 - 2k = 5$$

$$6 = 2k$$

$$k = 3$$

$\frac{1}{2}$ mark for solving for k

$\therefore$ The fourth term contains x^5 .

1 mark for fourth term
(or a term consistent with the value of k)

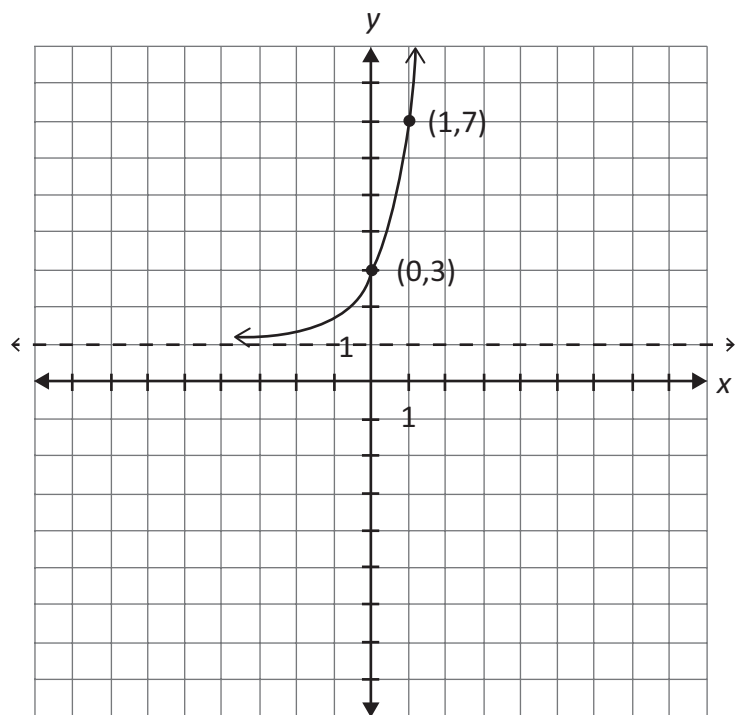
2 marks

Question 39

R9

Sketch the graph of $y = 2(3^x) + 1$.

Solution



1 mark for shape of an increasing exponential function

1 mark for vertical stretch

1 mark for vertical translation

3 marks

Question 40

R8, R10

Solve, algebraically.

$$\log(2x+1) = \log(x-2) + 1$$

Solution

Method 1

$$\log(2x+1) - \log(x-2) = 1$$

$$\log\left(\frac{2x+1}{x-2}\right) = 1$$

$$10^1 = \frac{2x+1}{x-2}$$

$$10x - 20 = 2x + 1$$

$$8x = 21$$

$$x = \frac{21}{8}$$

1 mark for quotient law

1 mark for exponential form

2 marks

Method 2

$$\log(2x+1) = \log(x-2) + \log 10$$

$$\log(2x+1) = \log(10(x-2))$$

$$2x+1 = 10(x-2)$$

$$2x+1 = 10x-20$$

$$21 = 8x$$

$$x = \frac{21}{8}$$

½ mark for logarithmic form

1 mark for product law

½ mark for equating arguments

2 marks

Question 41

R1

Given the functions, $f(x) = x + 2$ and $g(x) = x - 2$,

a) state the equation of $h(x) = f(x) \cdot g(x)$.

b) sketch the graph of $h(x) = f(x) \cdot g(x)$.

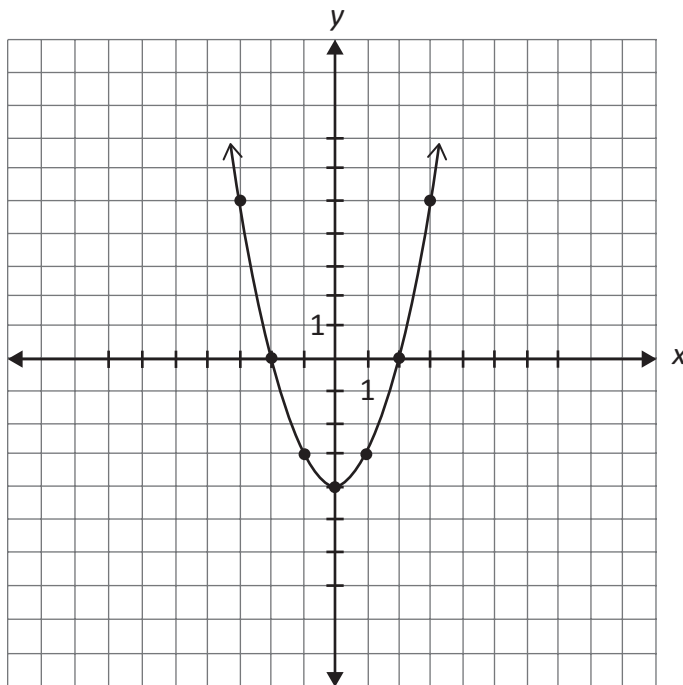
Solution

a) $h(x) = (x + 2)(x - 2)$

or

$$h(x) = x^2 - 4$$

b)



1 mark

1 mark for graph consistent with a)

1 mark

Question 42

R11

Given $p(x) = -2x^3 + 3x + 1$, determine the remainder when $p(x)$ is divided by $(x - 2)$.

Solution

Method 1

$$p(2) = -2(2)^3 + 3(2) + 1$$

$$p(2) = -2(8) + 6 + 1$$

$$p(2) = -9$$

1 mark for remainder theorem

1 mark

Method 2

2		-2	0	3	1
		↓	-4	-8	-10
		-2	-4	-5	-9

1 mark for synthetic division (or any equivalent strategy)

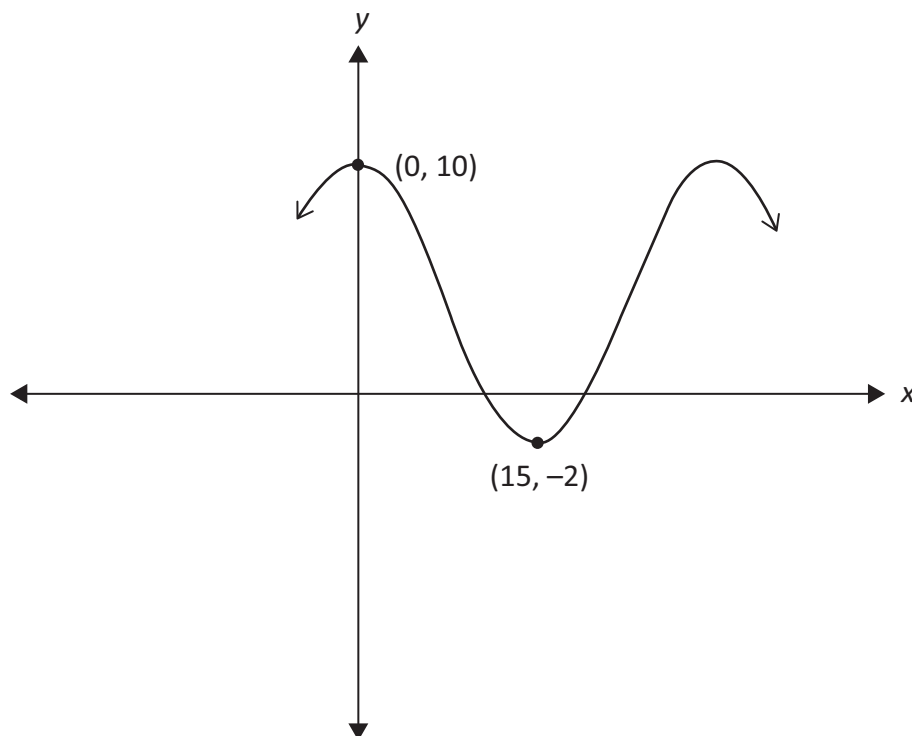
1 mark

∴ The remainder is -9

Question 43

T4

Determine the equation of the sinusoidal function in the form $y = A\cos(Bx) + D$.



Solution

$$y = 6\cos\left(\frac{\pi}{15}x\right) + 4$$

1 mark for the value of A

1 mark for the value of B ($\frac{1}{2}$ mark for period; $\frac{1}{2}$ mark for consistent value of B)

1 mark for the value of D

3 marks

Question 44

P2

Determine the total number of arrangements of the letters in the word TORONTO.

State your answer as a whole number.

Solution

$$\frac{7!}{2!3!}$$
$$\frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot \cancel{3!}}{2! \cancel{3!}}$$
$$\frac{7 \cdot 6 \cdot 5 \cdot 4}{2}$$
$$7 \cdot 6 \cdot 5 \cdot 2$$
$$420$$

½ mark for 7!

½ mark for division by 2!3!

½ mark for factorial expansion

½ mark for consistent number of arrangements

2 marks